

```

#include<cstdio>
#include<cstdlib>
#include<cstring>
#include<iostream>
#include<algorithm>
#define For(i,x,y) for (i=x;i<=y;i++)
using namespace std;
const int N=105;
const int N1=N*N;
const int T=63;
int i,j,k,n,m,x,p,t,tt,nn;
int d[N1],e[N1],h[N1],f[N1],hh[N1],re[N1];
char P[N1][10];
inline int C(int x,int y) { return !x||!y||x>n||y>m?0:(x-1)*m+y; }
inline void M(int &x,int y) { x+=y; if (x>=p) x-=p; }
inline void prepare()
{
    int i,j,k,v;
    For(i,1,nn) e[i]=i;
    For(k,1,T)
    {
        For(i,1,nn) hh[i]=h[i],h[i]=0;
        For(i,1,n) For(j,1,m)
        {
            v=C(i,j); int A=(k-1)%d[v]+1; f[v]=v;
            if (P[v][A]=='U') f[v]=C(i-1,j);
            if (P[v][A]=='D') f[v]=C(i+1,j);
            if (P[v][A]=='L') f[v]=C(i,j-1);
            if (P[v][A]=='R') f[v]=C(i,j+1);
            M(h[f[v]],hh[v]);
            if (P[v][A]=='C') M(h[v],x);
        }
        For(i,1,n)For(j,1,m) v=C(i,j),e[v]=f[e[v]];
    }
}
inline void mul()
{
    int i;
    For(i,1,nn) hh[i]=re[i],re[i]=h[i];
    For(i,1,nn) M(re[e[i]],hh[i]);
}
inline void multiply()
{
    int i;
    For(i,1,nn) hh[i]=h[i];
    For(i,1,nn) M(h[e[i]],hh[i]);
    For(i,1,nn) f[i]=e[i];
    For(i,1,nn) e[i]=f[f[i]];
}
inline void work1() { for (;t;t>>=1) { if (t&1) mul(); multiply(); } }
inline void work2()
{
    int i,j,k,v;
    For(k,1,tt)
    {
        For(i,1,nn) hh[i]=re[i],re[i]=0;
        For(i,1,n) For(j,1,m)
        {
            v=C(i,j); int A=(k-1)%d[v]+1; f[v]=v;
            if (P[v][A]=='U') f[v]=C(i-1,j);
            if (P[v][A]=='D') f[v]=C(i+1,j);
            if (P[v][A]=='L') f[v]=C(i,j-1);
            if (P[v][A]=='R') f[v]=C(i,j+1);
            M(re[f[v]],hh[v]);
            if (P[v][A]=='C') M(re[v],x);
        }
    }
}
inline void print()
{
    int i,j,k;
    For(i,1,n) for (j=1;j<=m;j=k)
    {
        for (k=j;k<=m&&re[C(i,j)]=re[C(i,k)];k++);
        printf("%d %d%c",k-j,re[C(i,j)],k>m?' \n':' ');
    }
}
int main()
{
    freopen("glob.in","r",stdin);
    freopen("glob.out","w",stdout);
    scanf("%d%d%d%d%d",&n,&m,&x,&p,&t),nn=n*m,x%=p;
    For(i,1,n) For(j,1,m) scanf("%d%s",&d[C(i,j)],P[C(i,j)]+1);
    tt=t%T,t/=T,prepare();
    work1(),work2(),print();
    return 0;
}

```